

WASP-163b

R-2119

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_260325 · created 2026-08-11T00:59:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461124.957 +/- 0.018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.023
Transit depth (Rp/Rs)^2	2.21 +/- 0.68 [%]
Orbital Inclination (inc)	89.65 +/- 2.65
Airmass coefficient 1 (a1)	1.044 +/- 0.013
Airmass coefficient 2 (a2)	-0.035 +/- 0.064
Scatter in the residuals of the lightcurve fit is	1.3 %
Best Comparison Star	#3 - [407, 49]
Optimal Aperture	2.84
Optimal Annulus	13.03
Transit Duration (day)	0.063 +/- 0.017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.149 ± 0.023 vs 0.119 (deviation 1.3σ)
Duration fit vs published	0.063 d vs 0.0942 d (deviation 1.83σ)
Mid-time O–C	-7.2 min
Depth SNR	6.5
Residual scatter	1.3 %

Light curve

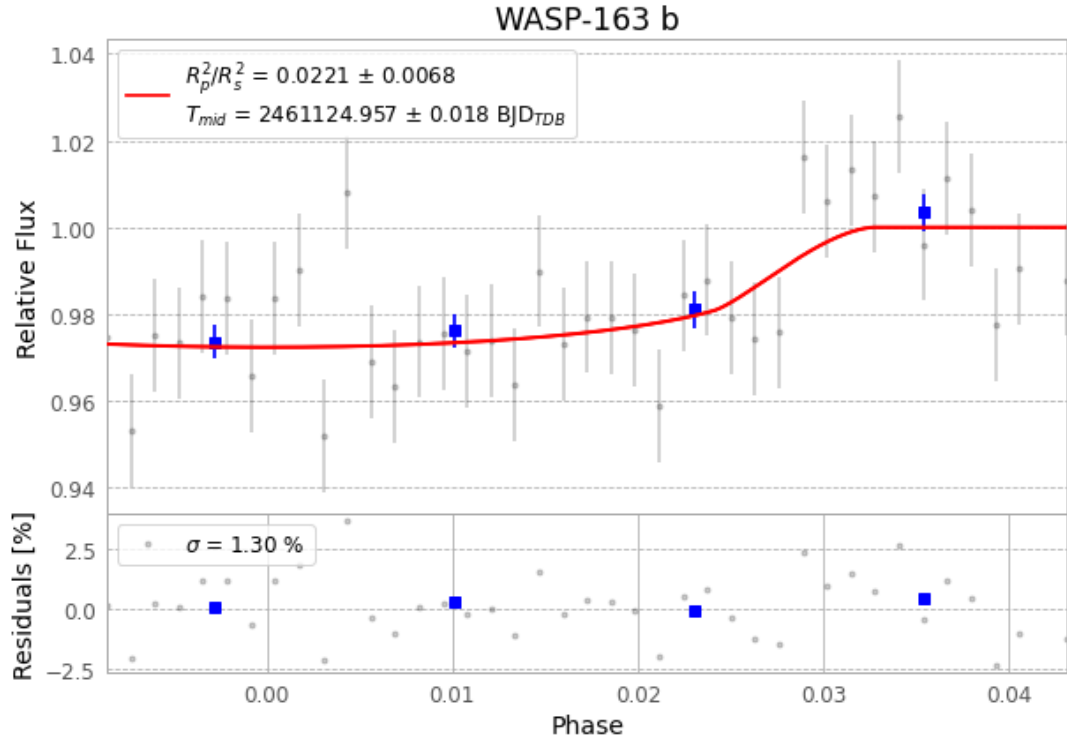


Figure 1 — FinalLightCurve_WASP-163 b_2026-03-25.png (SHA-256 3c11fd07e68bc16a...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 162], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 163e3e6a43124606...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

44 FITS frames (29 MB), WASP-163260325103315.FITS → WASP-163260325125720.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260325.log (4c2630365182...), FinalLightCurve_WASP-163 b_2026-03-25.png (3c11fd07e68b...), FinalLightCurve_WASP-163 b_2026-03-25.csv (4ae727f04b8f...)

Compute resources

Wall time 116.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-11 00:59Z.